

```

=====
N = 10 (h = 0.2000)
=====
t      Exact      Euler Taylor 2nd      Midpoint      Heun's
3rd      RK4      Adams-Bashforth      Adams-Moulton      Pred-Corrector
0.00000000 1.00000000 1.00000000 1.00000000 1.00000000
1.00000000 1.00000000 1.00000000 1.00000000 1.00000000
0.20000000 1.49182470 1.40000000 1.44000000 1.48000000
1.49066667 1.49173333 1.49173333 1.49173333 1.49173333
0.40000000 2.22554093 1.96000000 2.07567299 2.19040000
2.22208711 2.22526834 2.22526834 2.22526834 2.22526834
0.60000000 3.32011692 2.74400000 2.99496382 3.24179200
3.31239119 3.31950696 3.31950696 3.31764964 3.31950696
0.80000000 4.95303242 3.84160000 4.32575403 4.79785216
4.93767113 4.95181918 4.94411002 4.94611091 4.95172385
1.00000000 7.38905610 5.37824000 6.25417693 7.10082120
7.36042176 7.38679372 7.36051784 7.37393550 7.38649862
1.20000000 11.02317638 7.52953600 9.05140995 10.50921537
10.97193538 11.01912642 10.95919005 10.99347860
11.01846292
1.40000000 16.44464677 10.54135040 13.11290098 15.55363875
16.35549833 16.43759819 16.31821352 16.38969714
16.43627512
1.60000000 24.53253020 14.75789056 19.01584725 23.01938535
24.38059618 24.52051314 24.29740853 24.43468346
24.51804179
1.80000000 36.59823444 20.66104678 27.60348736 34.06869032
36.34334204 36.57806681 36.17787924 36.42860207
36.57363785
2.00000000 54.59815003 28.92546550 40.10881168 50.42166167
54.17580854 54.56472152 53.86750332 54.30981136
54.55700735
max |error| 25.67268454 14.48933836 4.17648836
0.42234150 0.03342851 0.73064672 0.28833867 0.04114268
=====

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=====
N = 20 (h = 0.1000)
=====
t      Exact      Euler Taylor 2nd      Midpoint      Heun's
3rd      RK4      Adams-Bashforth      Adams-Moulton      Pred-Corrector
0.00000000 1.00000000 1.00000000 1.00000000 1.00000000
1.00000000 1.00000000 1.00000000 1.00000000 1.00000000
0.10000000 1.22140276 1.20000000 1.21000000 1.22000000
1.22133333 1.22140000 1.22140000 1.22140000 1.22140000
0.20000000 1.49182470 1.44000000 1.46421403 1.48840000
1.49165511 1.49181796 1.49181796 1.49181796 1.49181796
0.30000000 1.82211880 1.72800000 1.77197508 1.81584800
1.82180811 1.82210646 1.82210646 1.82206225 1.82210646
0.40000000 2.22554093 2.07360000 2.14459128 2.21533456
2.22503497 2.22552083 2.22535975 2.22541104 2.22552788

```


1.00000000	1.00000000	1.00000000	1.00000000	1.00000000
0.05000000	1.10517092	1.10000000	1.10250000	1.10500000
1.10516667	1.10517083	1.10517083	1.10517083	1.10517083
0.10000000	1.22140276	1.21000000	1.21551293	1.22102500
1.22139336	1.22140257	1.22140257	1.22140257	1.22140257
0.15000000	1.34985881	1.33100000	1.34011773	1.34923262
1.34984323	1.34985850	1.34985850	1.34985729	1.34985850
0.20000000	1.49182470	1.46410000	1.47750415	1.49090205
1.49180174	1.49182424	1.49182011	1.49182155	1.49182454
0.25000000	1.64872127	1.61051000	1.62898412	1.64744677
1.64868956	1.64872064	1.64871099	1.64871617	1.64872131
0.30000000	1.82211880	1.77156100	1.79600434	1.82042868
1.82207674	1.82211796	1.82210207	1.82211137	1.82211907
0.35000000	2.01375271	1.94871710	1.98016007	2.01157369
2.01369848	2.01375163	2.01372825	2.01374251	2.01375327
0.40000000	2.22554093	2.14358881	2.18321046	2.22278892
2.22547244	2.22553956	2.22550724	2.22552747	2.22554183
0.45000000	2.45960311	2.35794769	2.40709536	2.45618176
2.45951796	2.45960141	2.45955853	2.45958582	2.45960442
0.50000000	2.71828183	2.59374246	2.65395390	2.71408085
2.71817726	2.71827974	2.71822444	2.71826005	2.71828362
0.55000000	3.00416602	2.85311671	2.92614499	2.99905934
3.00403890	3.00416349	3.00409363	3.00413900	3.00416838
0.60000000	3.32011692	3.13842838	3.22626991	3.31396057
3.31996366	3.32011387	3.32002699	3.32008379	3.32011995
0.65000000	3.66929667	3.45227121	3.55719719	3.66192643
3.66911317	3.66929301	3.66918632	3.66925644	3.66930048
0.70000000	4.05519997	3.79749834	3.92209015	4.04642870
4.05498158	4.05519561	4.05506590	4.05515152	4.05520470
0.75000000	4.48168907	4.17724817	4.32443717	4.47130371
4.48143047	4.48168392	4.48152751	4.48163112	4.48169487
0.80000000	4.95303242	4.59497299	4.76808511	4.94079060
4.95272758	4.95302635	4.95283908	4.95296351	4.95303947
0.85000000	5.47394739	5.05447028	5.25727620	5.45957362
5.47358943	5.47394026	5.47371736	5.47386585	5.47395587
0.90000000	6.04964746	5.55991731	5.79668869	6.03282885
6.04922858	6.04963911	6.04937517	6.04955139	6.04965760
0.95000000	6.68589444	6.11590904	6.39148167	6.66627588
6.68540579	6.68588470	6.68557354	6.68578169	6.68590650
1.00000000	7.38905610	6.72749995	7.04734458	7.36623484
7.38848763	7.38904477	7.38867937	7.38892422	7.38907036
1.05000000	8.16616991	7.40024994	7.77055168	8.13968950
8.16551025	8.16615676	8.16572917	8.16601614	8.16618672
1.10000000	9.02501350	8.14027494	8.56802227	8.99435690
9.02424974	9.02499828	9.02449944	9.02483468	9.02503322
1.15000000	9.97418245	8.95430243	9.44738703	9.93876437
9.97330000	9.97416486	9.97358454	9.97397502	9.97420552
1.20000000	11.02317638	9.84973268	10.41706119	10.98233463
11.02215872	11.02315610	11.02248265	11.02293629	
11.02320327				

1.25000000	12.18249396	10.83470594	11.48632525	12.13547977
12.18132241	12.18247061	12.18169088	12.18221663	
12.18252523				
1.30000000	13.46373804	11.91817654	12.66541401	13.40970514
13.46239149	13.46371119	13.46281028	13.46341830	
13.46377430				
1.35000000	14.87973172	13.10999419	13.96561475	14.81772418
14.87818633	14.87970092	14.87866195	14.87936373	
14.87977370				
1.40000000	16.44464677	14.42099361	15.39937556	16.37358522
16.44287559	16.44461146	16.44341536	16.44422390	
16.44469525				
1.45000000	18.17414537	15.86309297	16.98042473	18.09281167
18.17211800	18.17410496	18.17273016	18.17366016	
18.17420126				
1.50000000	20.08553692	17.44940227	18.72390257	19.99255690
20.08321908	20.08549072	20.08391288	20.08498093	
20.08560125				
1.55000000	22.19795128	19.19434250	20.64650667	22.09177537
22.19530429	22.19789852	22.19609013	22.19731498	
22.19802519				
1.60000000	24.53253020	21.11377675	22.76665221	24.41141178
24.52951045	24.53247000	24.53040003	24.53180285	
24.53261500				
1.65000000	27.11263892	23.22515442	25.10464876	26.97461002
27.10919730	27.11257032	27.11020373	27.11180842	
27.11273609				
1.70000000	29.96410005	25.54766986	27.68289523	29.80694407
29.96018122	29.96402193	29.96131924	29.96315273	
29.96421125				
1.75000000	33.11545196	28.10243685	30.52609501	32.93667320
33.11099361	33.11536309	33.11227977	33.11437245	
33.11557907				
1.80000000	36.59823444	30.91268053	33.66149314	36.39502389
36.59316644	36.59813342	36.59461931	36.59700541	
36.59837958				
1.85000000	40.44730436	34.00394859	37.11913804	40.21650140
40.44154778	40.44718961	40.44318821	40.44590629	
40.44746990				
1.90000000	44.70118449	37.40434344	40.93217010	44.43923404
44.69465055	44.70105424	44.69650192	44.69959543	
44.70137313				
1.95000000	49.40244911	41.14477779	45.13714007	49.10535362
49.39503797	49.40230137	49.39712649	49.40064433	
49.40266386				
2.00000000	54.59815003	45.25925557	49.77436020	54.26141575
54.58974946	54.59798257	54.59210455	54.59610176	
54.59839432				
max error		9.33889446	4.82378983	0.33673429
0.00840057	0.00016746	0.00604549	0.00204828	0.00024429

Error table | $h = 0.2$ ($N = 10$)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	1.000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00		0.00e+00			
0.20000000	1.491825	9.18e-02	5.18e-02	1.18e-02	1.16e-03	9.14e-05
9.14e-05	9.14e-05		9.14e-05			
0.40000000	2.225541	2.66e-01	1.50e-01	3.51e-02	3.45e-03	2.73e-04
2.73e-04	2.73e-04		2.73e-04			
0.60000000	3.320117	5.76e-01	3.25e-01	7.83e-02	7.73e-03	6.10e-04
6.10e-04	2.47e-03		6.10e-04			
0.80000000	4.953032	1.11e+00	6.27e-01	1.55e-01	1.54e-02	1.21e-03
8.92e-03	6.92e-03		1.31e-03			
1.00000000	7.389056	2.01e+00	1.13e+00	2.88e-01	2.86e-02	2.26e-03
2.85e-02	1.51e-02		2.56e-03			
1.20000000	11.023176	3.49e+00	1.97e+00	5.14e-01	5.12e-02	4.05e-03
6.40e-02	2.97e-02		4.71e-03			
1.40000000	16.444647	5.90e+00	3.33e+00	8.91e-01	8.91e-02	7.05e-03
1.26e-01	5.49e-02		8.37e-03			
1.60000000	24.532530	9.77e+00	5.52e+00	1.51e+00	1.52e-01	1.20e-02
2.35e-01	9.78e-02		1.45e-02			
1.80000000	36.598234	1.59e+01	8.99e+00	2.53e+00	2.55e-01	2.02e-02
4.20e-01	1.70e-01		2.46e-02			
2.00000000	54.598150	2.57e+01	1.45e+01	4.18e+00	4.22e-01	3.34e-02
7.31e-01	2.88e-01		4.11e-02			

Error table | $h = 0.1$ ($N = 20$)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	1.000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00		0.00e+00			
0.10000000	1.221403	2.14e-02	1.14e-02	1.40e-03	6.94e-05	2.76e-06
2.76e-06	2.76e-06		2.76e-06			
0.20000000	1.491825	5.18e-02	2.76e-02	3.42e-03	1.70e-04	6.74e-06
6.74e-06	6.74e-06		6.74e-06			
0.30000000	1.822119	9.41e-02	5.01e-02	6.27e-03	3.11e-04	1.23e-05
1.23e-05	5.66e-05		1.23e-05			
0.40000000	2.225541	1.52e-01	8.09e-02	1.02e-02	5.06e-04	2.01e-05
1.81e-04	1.30e-04		1.31e-05			
0.50000000	2.718282	2.30e-01	1.23e-01	1.56e-02	7.72e-04	3.07e-05
4.62e-04	2.32e-04		1.31e-05			
0.60000000	3.320117	3.34e-01	1.78e-01	2.28e-02	1.13e-03	4.50e-05
8.36e-04	3.74e-04		1.28e-05			
0.70000000	4.055200	4.72e-01	2.51e-01	3.25e-02	1.61e-03	6.41e-05
1.35e-03	5.67e-04		1.16e-05			

0.80000000	4.953032	6.53e-01	3.48e-01	4.53e-02	2.25e-03	8.95e-05
2.05e-03	8.27e-04		9.19e-06			
0.90000000	6.049647	8.90e-01	4.74e-01	6.22e-02	3.09e-03	1.23e-04
3.00e-03	1.17e-03		5.22e-06			
1.00000000	7.389056	1.20e+00	6.38e-01	8.44e-02	4.20e-03	1.67e-04
4.27e-03	1.63e-03		9.77e-07			
1.10000000	9.025013	1.59e+00	8.50e-01	1.13e-01	5.64e-03	2.24e-04
5.96e-03	2.24e-03		1.02e-05			
1.20000000	11.023176	2.11e+00	1.12e+00	1.51e-01	7.52e-03	2.99e-04
8.18e-03	3.04e-03		2.34e-05			
1.30000000	13.463738	2.76e+00	1.47e+00	2.00e-01	9.95e-03	3.95e-04
1.11e-02	4.08e-03		4.19e-05			
1.40000000	16.444647	3.61e+00	1.92e+00	2.62e-01	1.31e-02	5.20e-04
1.49e-02	5.42e-03		6.76e-05			
1.50000000	20.085537	4.68e+00	2.49e+00	3.43e-01	1.71e-02	6.80e-04
1.98e-02	7.17e-03		1.03e-04			
1.60000000	24.532530	6.04e+00	3.22e+00	4.47e-01	2.23e-02	8.86e-04
2.62e-02	9.42e-03		1.50e-04			
1.70000000	29.964100	7.78e+00	4.14e+00	5.80e-01	2.89e-02	1.15e-03
3.45e-02	1.23e-02		2.13e-04			
1.80000000	36.598234	9.97e+00	5.31e+00	7.49e-01	3.74e-02	1.49e-03
4.52e-02	1.60e-02		2.96e-04			
1.90000000	44.701184	1.28e+01	6.79e+00	9.65e-01	4.83e-02	1.92e-03
5.88e-02	2.08e-02		4.06e-04			
2.00000000	54.598150	1.63e+01	8.66e+00	1.24e+00	6.20e-02	2.47e-03
7.63e-02	2.69e-02		5.50e-04			

Error table | h = 0.05 (N = 40)

t	Exact	Euler	Taylor 2nd	Midpoint	Heun's 3rd	RK4
Adams-Bashforth	Adams-Moulton	Pred-Corrector				
0.00000000	1.000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.00e+00	0.00e+00		0.00e+00			
0.05000000	1.105171	5.17e-03	2.67e-03	1.71e-04	4.25e-06	8.47e-08
8.47e-08	8.47e-08		8.47e-08			
0.10000000	1.221403	1.14e-02	5.89e-03	3.78e-04	9.40e-06	1.87e-07
1.87e-07	1.87e-07		1.87e-07			
0.15000000	1.349859	1.89e-02	9.74e-03	6.26e-04	1.56e-05	3.11e-07
3.11e-07	1.52e-06		3.11e-07			
0.20000000	1.491825	2.77e-02	1.43e-02	9.23e-04	2.30e-05	4.58e-07
4.59e-06	3.15e-06		1.57e-07			
0.25000000	1.648721	3.82e-02	1.97e-02	1.27e-03	3.17e-05	6.32e-07
1.03e-05	5.10e-06		3.76e-08			
0.30000000	1.822119	5.06e-02	2.61e-02	1.69e-03	4.21e-05	8.38e-07
1.67e-05	7.43e-06		2.73e-07			
0.35000000	2.013753	6.50e-02	3.36e-02	2.18e-03	5.42e-05	1.08e-06
2.45e-05	1.02e-05		5.58e-07			
0.40000000	2.225541	8.20e-02	4.23e-02	2.75e-03	6.85e-05	1.37e-06
3.37e-05	1.35e-05		9.00e-07			

0.45000000	2.459603	1.02e-01	5.25e-02	3.42e-03	8.52e-05	1.70e-06
4.46e-05	1.73e-05		1.31e-06			
0.50000000	2.718282	1.25e-01	6.43e-02	4.20e-03	1.05e-04	2.08e-06
5.74e-05	2.18e-05		1.79e-06			
0.55000000	3.004166	1.51e-01	7.80e-02	5.11e-03	1.27e-04	2.53e-06
7.24e-05	2.70e-05		2.36e-06			
0.60000000	3.320117	1.82e-01	9.38e-02	6.16e-03	1.53e-04	3.05e-06
8.99e-05	3.31e-05		3.03e-06			
0.65000000	3.669297	2.17e-01	1.12e-01	7.37e-03	1.83e-04	3.66e-06
1.10e-04	4.02e-05		3.82e-06			
0.70000000	4.055200	2.58e-01	1.33e-01	8.77e-03	2.18e-04	4.35e-06
1.34e-04	4.84e-05		4.73e-06			
0.75000000	4.481689	3.04e-01	1.57e-01	1.04e-02	2.59e-04	5.15e-06
1.62e-04	5.79e-05		5.80e-06			
0.80000000	4.953032	3.58e-01	1.85e-01	1.22e-02	3.05e-04	6.08e-06
1.93e-04	6.89e-05		7.04e-06			
0.85000000	5.473947	4.19e-01	2.17e-01	1.44e-02	3.58e-04	7.14e-06
2.30e-04	8.15e-05		8.48e-06			
0.90000000	6.049647	4.90e-01	2.53e-01	1.68e-02	4.19e-04	8.35e-06
2.72e-04	9.61e-05		1.01e-05			
0.95000000	6.685894	5.70e-01	2.94e-01	1.96e-02	4.89e-04	9.74e-06
3.21e-04	1.13e-04		1.21e-05			
1.00000000	7.389056	6.62e-01	3.42e-01	2.28e-02	5.68e-04	1.13e-05
3.77e-04	1.32e-04		1.43e-05			
1.05000000	8.166170	7.66e-01	3.96e-01	2.65e-02	6.60e-04	1.31e-05
4.41e-04	1.54e-04		1.68e-05			
1.10000000	9.025013	8.85e-01	4.57e-01	3.07e-02	7.64e-04	1.52e-05
5.14e-04	1.79e-04		1.97e-05			
1.15000000	9.974182	1.02e+00	5.27e-01	3.54e-02	8.82e-04	1.76e-05
5.98e-04	2.07e-04		2.31e-05			
1.20000000	11.023176	1.17e+00	6.06e-01	4.08e-02	1.02e-03	2.03e-05
6.94e-04	2.40e-04		2.69e-05			
1.25000000	12.182494	1.35e+00	6.96e-01	4.70e-02	1.17e-03	2.34e-05
8.03e-04	2.77e-04		3.13e-05			
1.30000000	13.463738	1.55e+00	7.98e-01	5.40e-02	1.35e-03	2.68e-05
9.28e-04	3.20e-04		3.63e-05			
1.35000000	14.879732	1.77e+00	9.14e-01	6.20e-02	1.55e-03	3.08e-05
1.07e-03	3.68e-04		4.20e-05			
1.40000000	16.444647	2.02e+00	1.05e+00	7.11e-02	1.77e-03	3.53e-05
1.23e-03	4.23e-04		4.85e-05			
1.45000000	18.174145	2.31e+00	1.19e+00	8.13e-02	2.03e-03	4.04e-05
1.42e-03	4.85e-04		5.59e-05			
1.50000000	20.085537	2.64e+00	1.36e+00	9.30e-02	2.32e-03	4.62e-05
1.62e-03	5.56e-04		6.43e-05			
1.55000000	22.197951	3.00e+00	1.55e+00	1.06e-01	2.65e-03	5.28e-05
1.86e-03	6.36e-04		7.39e-05			
1.60000000	24.532530	3.42e+00	1.77e+00	1.21e-01	3.02e-03	6.02e-05
2.13e-03	7.27e-04		8.48e-05			
1.65000000	27.112639	3.89e+00	2.01e+00	1.38e-01	3.44e-03	6.86e-05
2.44e-03	8.31e-04		9.72e-05			

1.70000000	29.964100	4.42e+00	2.28e+00	1.57e-01	3.92e-03	7.81e-05
2.78e-03	9.47e-04		1.11e-04			
1.75000000	33.115452	5.01e+00	2.59e+00	1.79e-01	4.46e-03	8.89e-05
3.17e-03	1.08e-03		1.27e-04			
1.80000000	36.598234	5.69e+00	2.94e+00	2.03e-01	5.07e-03	1.01e-04
3.62e-03	1.23e-03		1.45e-04			
1.85000000	40.447304	6.44e+00	3.33e+00	2.31e-01	5.76e-03	1.15e-04
4.12e-03	1.40e-03		1.66e-04			
1.90000000	44.701184	7.30e+00	3.77e+00	2.62e-01	6.53e-03	1.30e-04
4.68e-03	1.59e-03		1.89e-04			
1.95000000	49.402449	8.26e+00	4.27e+00	2.97e-01	7.41e-03	1.48e-04
5.32e-03	1.80e-03		2.15e-04			
2.00000000	54.598150	9.34e+00	4.82e+00	3.37e-01	8.40e-03	1.67e-04
6.05e-03	2.05e-03		2.44e-04			